

**The Green Virtues of Chinese Protectionism:
A comparative analysis of EU and US responses to the China garbage shock**

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Abstract

Can economic protectionism in the Global South induce environmental reforms in the Global North? In 2017, China enacted “Operation National Sword,” which banned the import of a subset of internationally traded waste and scrap products. This unexpected and unprecedented collapse in Chinese demand – justified in China on environmental grounds and also an initial move in the brewing US-China trade war – shook global markets. Both the European Union and United States have long relied on exports to achieve “not in my backyard” (NIMBY) waste management goals, and China had become the dominant importer of a variety of their post-industrial and post-consumer products such as waste plastics. Thus, after 2017 EU and US waste exporters had to adjust, by reducing exports and/or diverting to new foreign markets. Conceivably, the collapse in global demand for certain waste products could ripple back to EU and US waste management policy, providing a sufficient shock to advance reform. At the same time, it is well established that the EU has environmental bona fides far exceeding those of the US. We use novel quantitative data to trace the nature of the shock to exports, providing evidence that disruptions to exports manifested in ways more visible to EU audiences than those in the US. Qualitatively, we show that EU-level policy has made meaningful strides, both within the EU and internationally, whereas initially bipartisan US interest frayed, leaving the problem of adjustment to US municipalities. Our analysis departs from conventional policy debates on the links between free trade and the environment to show that protective measures – not just trade agreements – can foster new policy linkages under certain circumstances. As what we term the “China garbage shock” reinforced rather than challenge the preexisting environmental policy cleavage between the EU and US, Chinese protectionism bolstered trade-environment linkages in a most-likely case but did not overcome trade disconnects in a least-likely case.

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1. Introduction

We are all accidental exporters. Or more accurately, every citizen that is connected to a waste management value chain is potentially a direct supplier of commodities that may be traded internationally. And yet, the international dimension of our consumption habits gets lost to most citizens, who (if they were to think about it) probably expect that their waste is recycled domestically or sent to domestic landfills and/or incinerators. In the Global North, waste management policies and recycling or upcycling reforms are seen as a reflection of our community leaders' commitment (or lack thereof) to a green future. Waste management is therefore understood to be as local as the trashcan we take out at night and the garbage truck that picks it up in the morning, roaming from the Joneses' white picket fence to ours.

Individuals in the Global North are largely unaware that, far from an insignificant local residue from end users, waste and scrap are in fact a global, billion-dollar economy that spans continents and sectors ranging from plastic, chemical, metal, electronic, textile, paper, animal, and vegetable products sourced from industry and individual consumers. Inside the metaphoric black garbage bag of waste disposal and management that we would rather not touch, see, or smell, lies a complex political economy of local, national, and international stakeholders who play an invisible yet crucial role in determining the fate of our discarded materials.

This hidden international contest to acquire or get rid of these peculiar goods is called the waste trade. Waste products are not inferior in themselves and, when processed, can generate inputs useful to myriad industries including manufacturing, construction, heavy industry, and so on. Global waste product markets, like any other, respond to supply and demand. States around the world but especially in the Global North have long relied on international demand for waste products as a means of dealing with their domestic supply of waste. Both the European Union and United States have designed waste management policies around export, relying on foreign demand for waste products to achieve "not in my backyard" (NIMBY) goals. China became the dominant importer of a variety of EU and US post-industrial and post-consumer products. The waste trade has been instrumental in China's development and industrialization (Goldstein 2020).

Therefore, China's unexpected and unprecedented 2017 "Operation National Sword" policy upended the status quo: China banned the importation of a subset of internationally traded waste products, including several ash, residue, and slag products containing mixed metals; several yarn and textile waste products made of cotton, wool and animal hair, or artificial fibers; sorted and unsorted rags; unsorted paper; and all post-consumer plastics products. Thus, after 2017 EU and US waste exporters had to adjust, by keeping more waste products at home and/or finding new buyers in alternative foreign markets.

A large literature considers the political-economic consequences in the Global North of China's entry to global markets, particularly since China's accession to the WTO in 2001. This literature has summarized China's export boom and its resulting market penetration in the North as the "China shock" (Autor et al 2016). We use the 2017 "China garbage shock" to explore both

economic and political consequences when China exits markets in which it was an *importer*. Conceivably, the collapse in global demand for certain waste products could build trade linkages between Chinese protectionism and EU and US waste management policies. Has this happened?

In this article, we trace that while the China garbage shock opened opportunities for trade-environment linkages, it reinforced rather than challenged the preexisting environmental policy cleavage between the EU and US. As evidence, we examine trends in novel quantitative data and review discursive and policy responses in the EU as compared to the US. Quantitatively, while exports of banned products dropped sharply for the EU and US, trade diversion outcomes in the EU manifested in ways that were more visible to EU audiences than in the US, for which exports continued to flow to geographically and socio-economically far away destinations. Qualitatively, policymakers in the EU both discursively and in the policymaking process acknowledged the role of Chinese bans and used them as a springboard to deeper trade-environment linkages. In the US, despite initial bipartisan interest, policymakers remained discursively vague about the problem and federal-level efforts fizzled, leaving localities to bear the burden of adjustment. In short, the preexisting political economic context suggested more fertile ground for trade-environment linkages in the EU than the US, and our findings are that the China garbage shock grew those linkages in the EU most-likely case but stalled in the US least-likely case.

In what follows, we review limited social science work on this topic, introduce novel data on trade in waste, quantitatively compare trends between the EU and US, and qualitatively compare discourse and policy in the EU and US. We end by reinforcing the importance of this study, despite its unsurprising finding of more environmental progress in the EU as compared to the US, and the myriad opportunities it opens for future garbage research.

2. What is the global waste trade?

Companies buy waste and scrap because they can process it to recover raw materials, which then can be repurposed or sold onward as inputs into new production processes. A waste product contains raw materials that can be repurposed into inputs, and when a waste product is voluntarily bought and sold on international markets, we can use the fact that importers are paying (positive) money for the product as an indicator that there is value to be had from processing it.

To demonstrate what the global waste trade is all about, we leverage a new data collection effort (Wellhausen 2023) to provide insight into waste products. That data collection gathered unique waste products that have been classified by the World Customs Organization (WCO) and given official codes under the Harmonized System (HS), which allows observers to follow their import and export around the world. HS codes are hierarchical and move from categories (2- and 4-digit) to a specific product with a 6-digit HS code; codes beyond 6-digits are not globally harmonized. The primary sample is of 179 unique HS 6-digit waste products, over the period 1995-2020.¹

¹ See the appendix for the full list of HS codes.

To be clear, there are many ways wastes move around the world outside of legal international trade. The multiple oceanic Garbage Patches illustrate that plastic waste has been notoriously poorly accounted for (Lebreton et al 2018). Electronic waste (e-waste) has been largely unmeasured (Lepawsky 2018). The WCO itself acknowledges the e-waste measurement problem, calling e-waste an “example of a product class which presents significant policy concerns as well as a high value of trade.”² Moreover, standard international trade statistics cannot see trade in services, in which we can conceptualize that waste is exported at a negative price and the service of foreign waste disposal is imported. Illegal waste dumping – trade in services at a too-low price – is also a severe global problem but outside the scope here (e.g. Boidier and Bensebaa 2011, Favarin and Aziani 2020).

In short, the sample of legally defined waste products we examine is a best-available solution to the bigger problem of the poorly measured movement of waste around the world. We see it as crucial to advance research on the political economy of the international waste trade now, accepting data limitations. In fact, in explaining its 2017 policy, China itself acknowledged data issues around tracking the international movement of waste products. In a response to WTO complaints over its actions, China answered, “Currently, there is not any globally recognized standard for scrap materials and recyclable materials. China therefore adopts the universally applicable HS code.”³

To provide insight into specific waste products, Table 1 lists the sub-sample of 26 waste products China banned in its 2017 “Operation National Sword” policy. The first column of Table 1 lists the type of waste product, which per industry standards are categorized according to origin. While China banned products of animal, metal, paper, plastic, and textile origin, there are also traded waste products of chemical, mineral, and vegetable origin. The reader might be interested in additional examples of each. Animal waste products include hair-based products, such as those that China banned, as well as substances like feathers that can be processed into inputs like stuffing for furniture or animal byproducts such as raw fats that can be processed into inputs for a variety of production processes. Metal waste and scrap products are the backbone of many production processes in the world, including in the Global North; aluminum waste, for example, is very cost-effective and competitive as an alternative source of inputs compared to virgin aluminum produced from bauxite. Paper and plastic waste products are key sources of inputs into downstream production of a huge variety of consumer and industrial goods. Textile waste includes everything from broken thread and rags to the thriving international trade in bales of worn clothing. Chemical origin products include things like clinical waste and municipal waste (solid or ash). While these might be eyebrow-raising to outsiders, still there is demand for them on international markets. Mineral-origin wastes, like glass and different forms of broken rock, have a variety of industrial applications. Vegetable-origin waste includes things like residues from grain processing and seaweed ash that can be processed to generate feedstocks for animals.

² HS Nomenclature 2022 edition, Section XVI, New Note 6, (A)(ii). The WCO’s 2022 revision introduced the first e-waste product codes, which should help alleviate the problem.

³ Staub, Colin. 19 June 2018. “China: Waste restrictions will spur US job growth.” Resource Recycling.

Table 1. China banned imports of these 26 waste products (HS 6-digit) in its 2017 “Operation National Sword.”

Waste type (origin)	Short description	HS codes
Animal	Wool and fine animal hair waste	510310; 510320
Animal	Coarse animal hair waste	510330
Animal	Fine and coarse animal hair waste (garnetted stock)	510400
Metal	Slag and waste from iron and steel manufacturing	261900
Metal	Ash and residue containing mainly zinc	262011; 262019
Metal	Leaded gasoline sludges	262021
Metal	Lead residue (other than leaded gasoline sludges)	262029
Metal	Ash and residue containing mainly copper	262030
Metal	Ash and residue containing mainly aluminum	262040
Metal	Mixed residues and ash, non-ferrous including toxic	262060
Metal	Mixed residues and ash, non-ferrous and non-toxic	262091; 262099
Paper	Unsorted paper waste and scrap	470790
Plastic	Ethylene polymer waste, pairings, and scrap	391510
Plastic	Styrene polymer waste, pairings, and scrap	391520
Plastic	Vinyl chloride waste, pairings, and scrap	391530
Plastic	Plastic waste and scrap "not elsewhere specified"	391590
Textile	Cotton waste (yarn and thread)	520210
Textile	Cotton waste (garnetted stock)	520291
Textile	Cotton waste, "other than..."	520299
Textile	Synthetic fiber waste (unspecified)	550510; 550520
Textile	Rags, sorted	631010
Textile	Rags, unsorted	631090

Source: Chinese notification to the WTO, 18 July 2017 (G/TBT/N/CHN/1211).

What all the waste products in Table 1 as well as the additional dozens of traded waste products have in common is that the importer is knowingly, intentionally buying what is called end-of-life (EOL) content mixed in with the raw materials of value. Consider international trade in sewage sludge (HS 382520). An importer buys sewage, which may be better treated or have more consistent qualities than domestic sewage. The importer processes that sewage into fertilizer, then uses or sells the fertilizer onward. Importers of waste products are thus providing a service: processing out the junk and generating the desired input. (Alternatively, the importer could simply buy fertilizer, which is a primary good.) We can expect the exporter and importer to bargain over the price of the waste product, taking this service provision into account. And yet, because of that service provision, EOL content – foreign-origin sewage residue of no further use

– is left in the importer’s geography. Clearly, there are political, economic, social, and environmental consequences of waste product transactions that cannot be assumed to be fully internalized in the price. Put differently, there are many “not in my backyard” (NIMBY) benefits of getting the EOL content out of the exporter’s jurisdiction, and “yes in my backyard” (YIMBY) costs of accepting foreign-origin trash into the importer’s jurisdiction.

3. What we (do not) know about the politics of the global waste trade

What do social scientists have to say about waste trade? Given its inherent interdisciplinary nature, various bodies of literature have shed light on different aspects. One strand of research, drawing from environmental and trade economics, has sought to measure the waste trade and its economic ramifications in an international “circular economy” (Kellenberg 2015; Yamaguchi 2021). Another literature, which might be described as post-colonial, has shed light on the power struggle inherent to waste generation and trade, showing, in Liboiron’s terms (2021), that “*pollution is colonialism*” (see also Okafor-Yarwood and Adewumi 2020, Cotta 2020). Within international relations, scholars of environment have made great inroads on climate (e.g. Bernauer 2013), though waste management has received less attention (but see O’Neill 2000, 2019).

Within the field of environmental governance, scholars have approached the regulation of waste through the lens of multilateral environmental agreements (MEAs) (Roberts et al 2004, Morin et al 2018, Brandi et al 2019, Morin et al 2024). Multilateral treaty-making has indeed played an important role in the waste trade. The 1989 Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal is the most successful global governance effort to date and has near universal membership – with the United States as the major exception. Trade-in-services scandals in the 1980s between Global North exporters and Tonga, Western Samoa, Nigeria, and Guinea-Bissau helped motivate the Basel Convention, though there remains no outright ban on hazardous waste shipments intended for final disposal in Global South states (Yang 2020; O’Neill 2019; McKee 1996). A set of African nations formed the 1998 Bamako Convention that prohibits hazardous waste imports into member-states; however, there are serious concerns over compliance (Okafor-Yarwood and Adewumi 2020). Starting in 2022, the United Nations began negotiations for a Global Plastics Treaty, intended to be a legally binding instrument to reduce plastic pollution, although plans to complete negotiations in 2024 failed and the process has continued into 2025.

Finally, the enforcement of MEAs has received significant attention amidst trade-environment debates, with scholars mapping out the scope of environmental provisions in trade agreements, their enforceability, or the processes of policy diffusion through trade agreements (Bastiaens and Postnikov 2017; Jinnah and Morin 2020; Brandi, Blumer & Morin 2019; Brandi & Morin 2023; Velut et al 2022; George & Yamaguchi 2018). However, compared to other environmental issues like air pollution or climate, or specific MEAs like the Convention on International Trade of Endangered Species of Wild Flora and Fauna (CITES), waste management and trade has received much less attention.

This is because MEAs and Free Trade Agreements (FTAs) specifically provide little scope for waste trade. Admittedly, several EU trade and association agreements address hazardous waste trade by referencing the Basel Convention. On the US side, hazardous waste was mentioned in the unratified Trans-Pacific Partnership but otherwise has been absent from its FTAs. The subsequent CPTPP (of which the US is not a party) includes explicit reference to the Basel Convention (of which the US is not a party). Beyond these, most EU and US FTAs since the 2010s include domestic waste management provisions, typically in clauses referring to cooperation to protect countries “against the pollution of fresh and marine waters, air and soil” (EC-Central America FTA 2012; See also USMCA 2018). Scholar-activists have discussed waste trade-related treaty clauses that could shift more environmental burdens to Global North signatories, though there has not (yet) been uptake of these ideas (Torrente-Velásquez et al 2020). As is often the case with environmental provisions in FTAs, what waste management provisions exist remain primarily aspirational: to date, neither the EU nor the US has ever used dispute settlement mechanisms to enforce environmental provisions, let alone clauses pertaining to domestic waste management or waste trade (Velut et al 2022).

Another pitfall of the trade-environment literature pertains to its primary focus on FTA provisions. Yet, as the editors of this special issue note, the confluence of internal factors and external shocks to the international trading system has fractured the trade-liberalizing consensus. Trade policy tools have diversified far beyond formal FTAs, and they are being deployed in pursuit of a great variety of objectives (Velut, Dalingwater, Siles-Brügge and Ouellet, this issue; see also DeVille, Happersberger and Kalimo 2023). Following the editors’ call, we examine the realization of trade-environment linkages outside the scope of treaty design, to explore the extent to which protective measures can impact – or not – environmental reforms. At the same time, we shed light on a severely understudied topic area in international environmental politics: global trade in waste and scrap. China’s 2017 “Operation National Sword” policy gives us a notable opportunity to understand how unilateral non-tariff measures enacted by a key Global South trading partner affect outcomes in the Global North. Because the topic area is waste, we specifically expect the fallout of China’s trade policy to manifest in environmental policy in the EU and US. This is another, albeit secondary, contribution of this article: decentering the narrative of trade-environment linkages away from the Global North, and more specifically the US and the EU.

4. How did the China garbage shock affect EU and US waste trade flows?

China’s rise and its near-insatiable demand for raw materials over the last decades have been central in the deepening of global markets for waste and scrap (Minter 2015, Goldstein 2020). So, on 18 July 2017, the Chinese government truly shocked the waste trade when it notified to the WTO that it would be banning the import of 26 specific waste products (see again Table 1). The precise timing of the China garbage shock was a product of issue linkage, serving as one of China’s opening salvos in the US-China trade war. For its part, the global waste management industry had thought that China’s efforts to revamp regulatory enforcement of waste imports

four years earlier in its “Operation Green Fence” was successful enough to preclude the need for further action anytime soon, and certainly not at such a scale.

Post-“China garbage shock” material disruptions to status quo trade flows could occur through two main channels. First, exports could decline – waste products that were previously shipped to China could be kept at home and no longer sold abroad as waste products. Second, exports could divert – waste products that were previously shipped to China could be sold to importers in other countries. A set of important studies of the China garbage shock have traced trade declines and diversions globally, for plastic waste (Brooks et al 2018, Pacini et al 2021), for paper waste (Ma et al 2021), and for combinations of products (Tran et al 2021). Our empirical contribution to the literature lies in the use of a novel dataset to conduct a comparative analysis of the EU and US cases. By capturing the cumulative effects of China’s import bans, across all banned products, we aim to show that greater magnitude disruptions with characteristics more visible to societies encountering disruption are more likely to generate policy responses. This allows us to highlight outcomes specific to particular exporting countries and identify factors conducive to stronger trade-environment linkages.

Of course, foregone trade and trade diversion are likely deeply endogenous to regulatory and government responses as they unfold over time. Our strategy is to focus on aggregate data in the immediate aftermath of the China garbage shock, from 2018-2020, and compare it to relevant pre-periods. We are not interested in causality directly; rather, we want to capture the trends experienced by market and political actors in the EU and US.

4.1 General declines in exports of China ban waste products

In Figure 1, we plot worldwide trends in exports of the 26 products subject to the 2017 China ban, from 1995-2020.⁴ Exports of these waste products grew precipitously over the 1990s. In terms of market value, in US dollars these products fetched the highest prices in the early 2010s, and the aggregate value of this trade was already declining prior to 2017. In terms of volume, however, trade in these products was high and increasing in advance of 2017, with the absolute volume (in millions of metric tons) exported in 2016 having reached a new height. After 2017, exports of these China ban products plummeted, in terms of volume and in terms of price. In 2020, the world market for these products was much closer to what it had been in the mid-2000s.

⁴ All trade data are from BACI: CEPII/COMPUSTAT, version 202201 (Gaulier and Zignago 2010). On the importance of and considerations around analyzing product-level data, see Kim et al (2020).

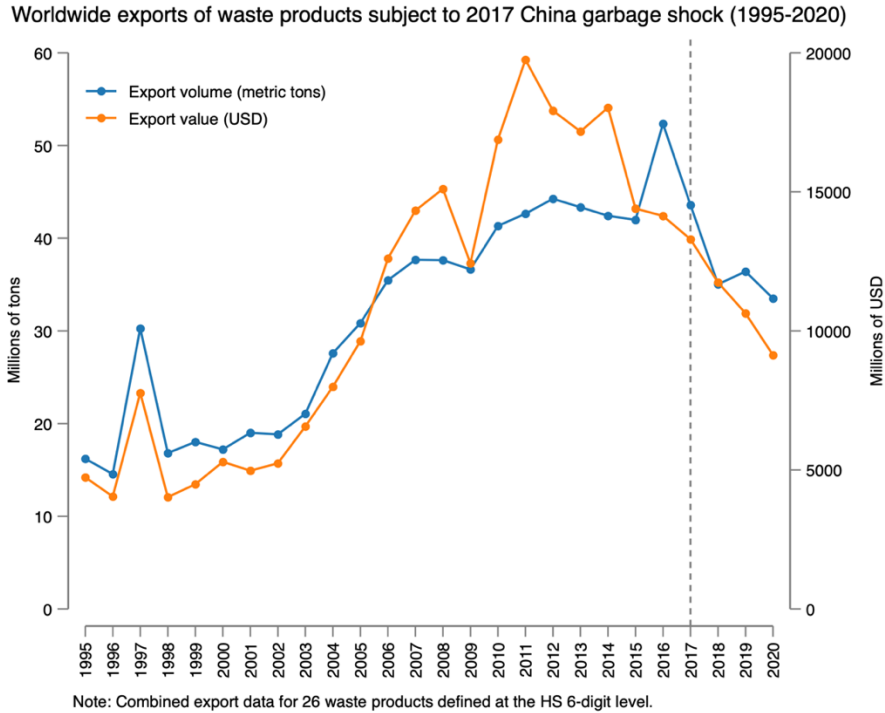
FIGURE 1

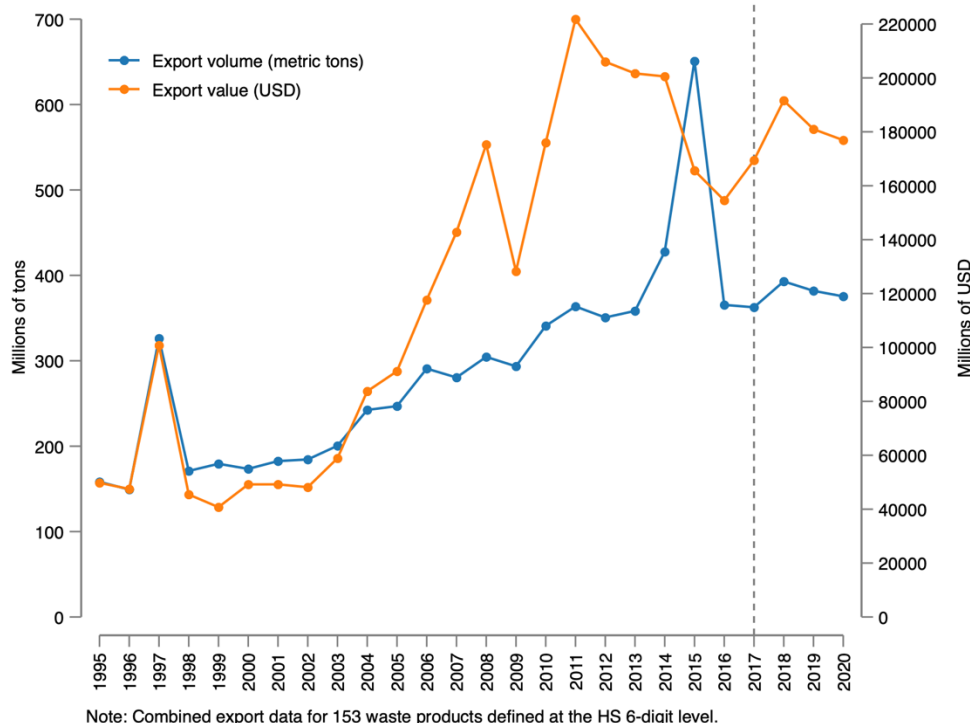
Figure 1 implies that China's closure of its markets for these specific waste products had global consequences. For added confirmation that it was the China garbage shock and not something else changing trends, Figure 2 plots trends using the dataset on the full spectrum of 179 waste products legally traded on international markets in the period (Wellhausen 2023).⁵ If trends in exports of the 153 other waste products look like those 26 aggregated in Figure 1, we would have less confidence that the 2017 China garbage shock itself had an effect on the global waste trade. However, in Figure 2 we can see clearly that the trends in the remainder of the waste trade are different. First, by comparing the y-axes in Figures 1 and 2, we can see that the amount of waste products affected by the China ban, in volume or value, is only a fraction of the global waste trade. Second, in Figure 2 we see that the trend after 2017 for the rest of the waste trade was fundamentally different. Export values had been declining in previous years, but they jumped back to values last seen in the early 2010s and the decline flattened out. Moreover, in the aftermath of the China garbage shock export volumes of other waste products continued longer-term secular increases and remained at some of their historically highest levels.⁶

FIGURE 2

⁵ See again the Appendix for the full list of HS codes.

⁶ In the industry, the spike in export volume in 2014/2015 is understood as resulting from the decommissioning of an unusually large number of ships; decommissioned ships trade as HS 890800 for ship salvage.

Worldwide exports of waste products *not* subject to 2017 China garbage shock (1995-2020)



Taken together, Figures 1 and 2 demonstrate that the China garbage shock did not decimate the global waste trade per se, but it did fundamentally decrease aggregate exports of the waste products it targeted. For the remainder of this article, we will focus in on only the 26 waste products whose import was banned by China, and the consequences for exporters from the EU-27 and the US.

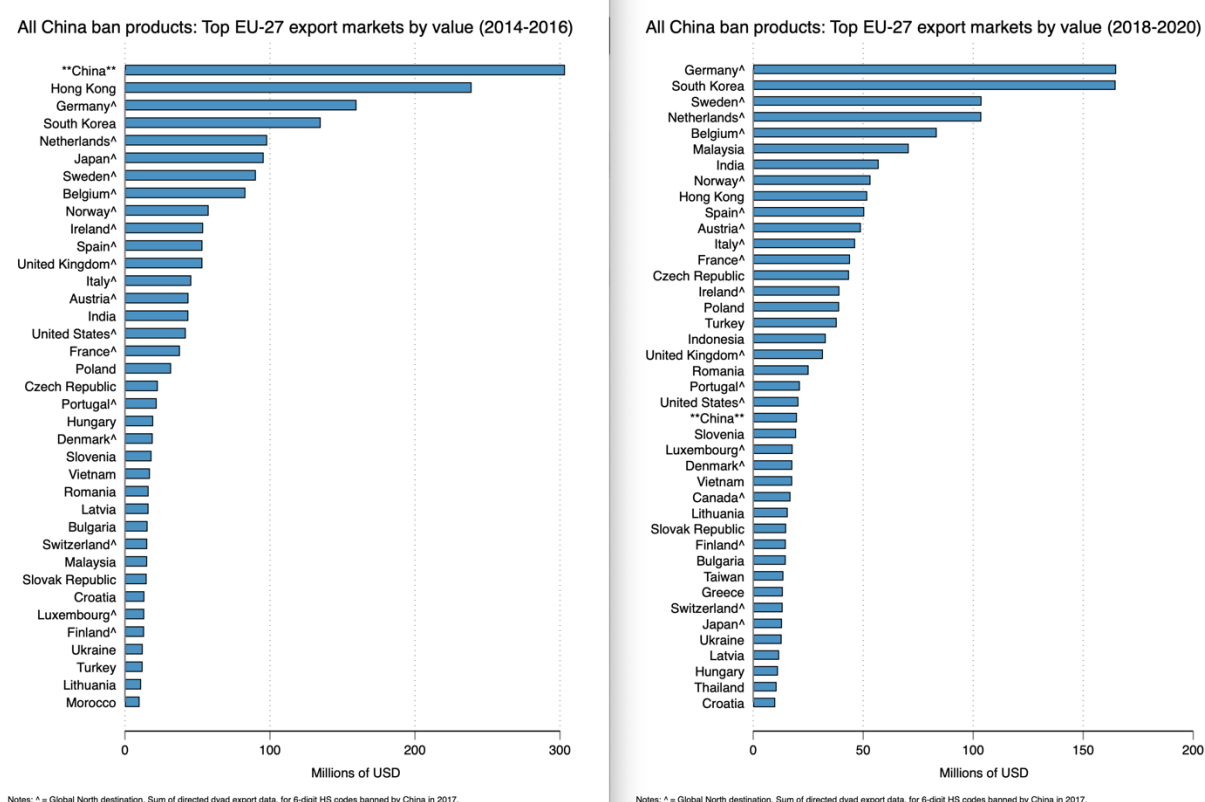
4.2 Variation in trade diversion by EU and US exporters

The general decline in exports of banned waste products shows up in global data, driven in no small part by the US and the countries of the EU. Via this channel, the China garbage shock had similar effects. Now we turn to the trade diversion channel, by looking at the top destinations of exports before and after the shock, for EU and US exporters. In sum, this channel reveals important divergence in the fallout of the China garbage shock. Exporters in both the EU and the US diverted to new markets. However, the EU's new markets were closer to home compared to the US's new markets which remained far removed. Environment is already more salient in the EU than the US, and these trade diversion trends reinforce that divergence.

To understand the trade diversion channel, we look at the top destinations of exports. We first focus on EU-27 exports in terms of value (USD millions), in the years immediately preceding the

China ban (Figure 3, left panel) and the years immediately after (Figure 3, right panel).⁷ Focusing on the pre-ban period (left), we see that China dominated as the primary destination. Notably, Hong Kong was second. Industry insiders acknowledge Hong Kong as a destination in itself but also as a waypoint for re-export to China or elsewhere in Southeast Asia. South Korea is also a major destination market. Many of the other top destinations of exports of China ban waste products, albeit in significantly lower values, are in fact in the Global North (marked with a ^ in Figure 3).⁸ Again, as waste products are bought and sold as substances from which to recover raw materials, it makes sense that there is demand in the Global North, too. Even when China was open as a market destination, exports of EU waste products to Global North states was not negligible.

FIGURE 3



The right panel of Figure 3 shows how the top EU-27 export destinations for China ban waste products changed in the immediate aftermath of the China garbage shock. China as a destination dropped precipitously, with export sales in the 2018-2020 period falling to a fraction of what they

⁷ We exclude consideration of the year 2017 itself, as the ban occurred in July. For simplicity, all statistics are always calculated with the EU-27 (members as of the time of writing) as the states included in the EU, even if and when this is historically anachronistic. We are happy to share disaggregated data with the interested reader.

⁸ Global North states are defined as (1) OECD members and (2) high income by the World Bank, as of the beginning of our trade data (1995). All other states are Global South. This means some EU-27 members are Global North and others are not.

were in the pre-period. That export sales to China did not fall to zero shows that there were limits on the enforcement of the bans. While China directed its regulators to fully implement import bans on unsorted waste paper and post-consumer plastics in the five months before the end of 2017, the implementation timeline for other bans was delegated to regulators, although the goal of speed was implied. Still, we can infer from the non-zero exports to China in the immediate post-period that it is politically difficult to eliminate a competitive domestic industry, perhaps especially one that is serving domestic demand for raw materials. In the post-period, exports to Hong Kong also dropped to less than a quarter of what they once were. We can infer from the right panel of Figure 3 that EU-27 exporters that needed to diversify away from China found buyers in many previously less important export destinations. From the data, it appears South Korea, Malaysia, India, Turkey, and non-Global North EU member states became important destinations that absorbed diverted exports. Still, Global North states in and out of the EU grew in importance as export destinations. That the China garbage shock led to more waste products flowing into such a variety of Global North states is notable, and notably different than the fallout in the US which we turn to next.

FIGURE 4

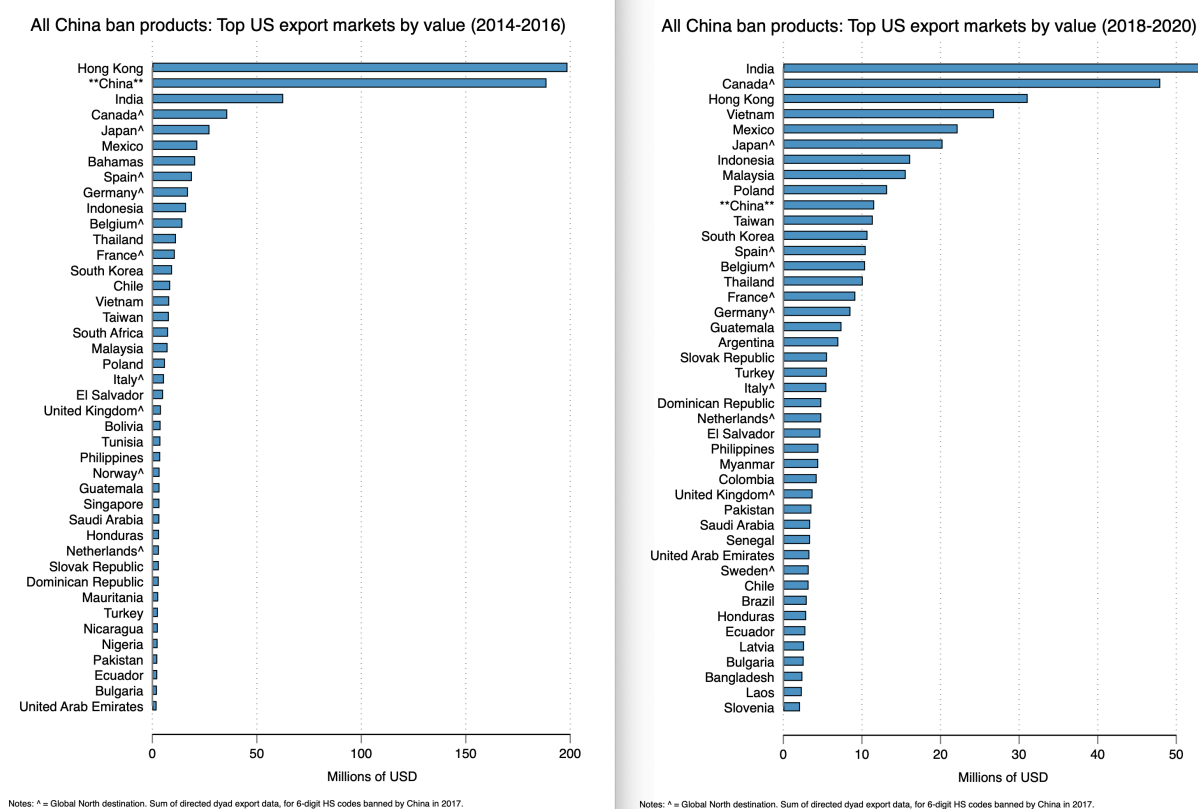


Figure 4 reproduces the pre- and post-China ban data for the value of exports of banned products, but for the US rather than the EU-27. The dominance of China in the pre-period, and its fall although not to zero in the post-period, parallels the EU data. Again, similar to the EU data,

US exports to Hong Kong were disrupted. However, US exports to other Global North states did not play as large a role in either the pre- or post-period as they did for the EU – perhaps in no small part due to its geographic location. Canada is the major exception, although it is not obvious Canada did much to absorb new exports as the pre- and post-period values are about on par (close to USD 50 million). A variety of Global South destinations became more important in the post-period, especially India, Vietnam, Mexico, Indonesia, and Malaysia.

Comparing across Figures 3 and 4, we see as a key axis of differentiation the prevalence of Global South rather than Global North buyers in the US post-period, in comparison to the EU. For the EU-27, buyers absorbing diverted waste products were not so far afield, whether conceptualized geographically, socio-economically, or politically. That the fallout was not so far removed from EU audiences likely provided a public awareness complement to the EU's long and deep interests in environmental regulation. This raises the possibility that ideational factors may have played a role in fostering trade and waste management interlinkages in the form of coordinated policy responses ("trade for" linkages). For the US, China waste ban products continued to be sold to buyers in less politically powerful, developing destination markets, including markets geographically far away. The data support the reality that public awareness of these issues in the general US population is low, which combined with a US legacy of more punctuated and limited interest in environmental regulation sets the stage for variation on the policy side.

4.3 Case study: Fallout for plastic waste exports

One difficulty in drawing conclusions from the previous figures is that they combine markets for 26 different waste products banned by China, with very different physical characteristics, prices on world markets, downstream uses, and environmental consequences. While the China garbage shock applies to all 26 waste products, in terms of public policy post-consumer waste products suggest different regulatory tools and approaches compared to waste products resulting from industrial production. Of special importance are post-consumer plastic waste products, which China banned in full.

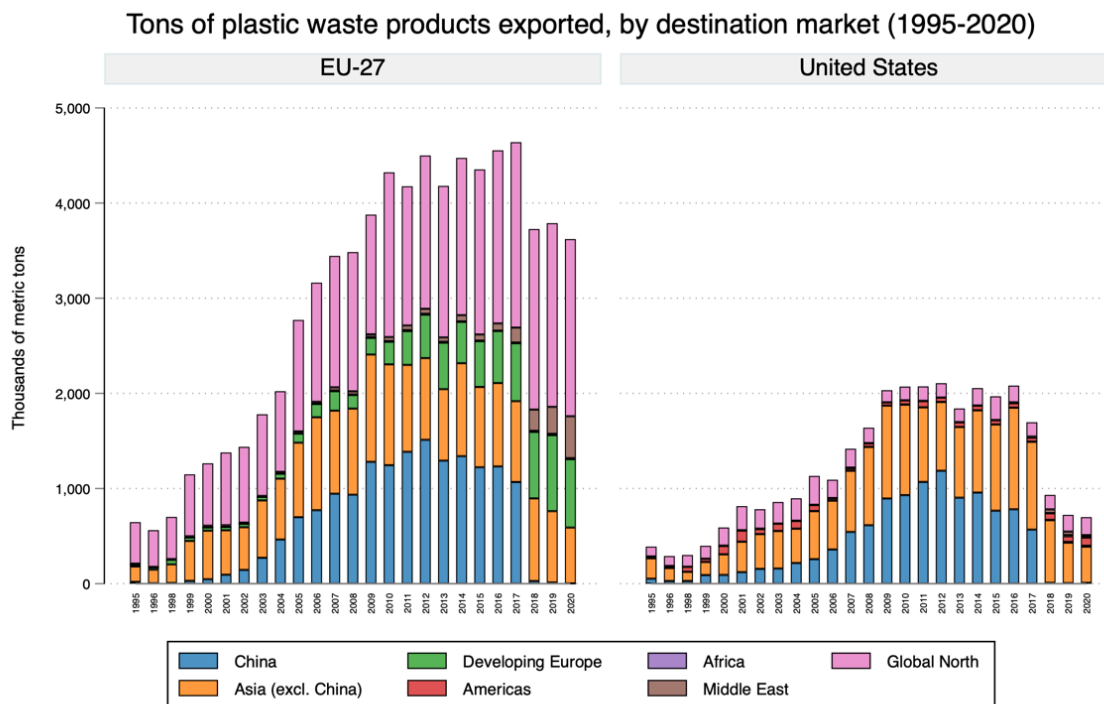
In zooming in on waste plastics, we pivot from examining financial data on the value of trade, to data on trade volume, or the literal weight of plastic waste products exported. Why focus on volume? First, the physicality of waste matters for the environmental, political, social, economic, and even moral consequences of its movement around the world. Whatever the dollar value attached to the trade, the export of a waste product from one jurisdiction to another also moves physical, polluting, even smelly substances. Examining the physical footprint of plastic waste gets us more direct insight into the externalities that imports might generate.

Second, if global markets for a given product decline, then usually what would happen is that would-be exporters would just reduce production of the product. Plastic waste – like waste products in general – is unusual, in that its production is highly inelastic. Whatever happens with downstream markets for plastic recyclables, you and I are still consuming plastics and, in turn, producing plastic waste. We might be trying to adjust our consumption habits, and regulatory policies can push us to do so. But certainly, in the years immediately after the China garbage

shock, the consumption of primary plastic goods in the EU or the US did not stop, so the production of physical plastic waste did not stop. What happened is that commercial markets in China were no longer a geographic destination for millions of tons of that physical plastic waste.

Figure 5 uses data on the physical volume of plastic waste exports to show us what happened. The data underlying Figure 5 are export volumes for the four post-consumer plastic products banned by China in 2017 (see again Table 1). The left panel shows data for exports out of EU-27, and the right panel for exports out of the US. Each bar captures the total volume of exports in millions of metric tons of plastic waste products in the noted year. Bars are subdivided to show where the importers were located. To be clear, some portion of these millions of tons of plastic waste is repurposed into new inputs for downstream uses; these are raw materials of value, and importing these materials should not be interpreted as polluting per se. However, because these are traded waste products, by definition some portion of the volume recorded in Figure 5 becomes EOL waste with no further value. We assume a positive and large-in-magnitude relationship between the quantity of waste product imported and the quantity of EOL waste left over after processing, destined to be stored forever in the importers' jurisdiction or transformed into air and effluent pollution that seeps beyond the importers' geographic boundaries (Kaza et al 2018).

FIGURE 5



Notes: Sum of directed dyad export data for plastic waste 6-digit HS codes. China banned their import in 2017. Global North states are, as of 1995, (1) in the OECD and (2) in the World Bank's High Income classification group.

In Figure 5, plastic waste exported to China is represented in blue (lowest bar). We can see the role of China paralleled for both the EU-27 (left panel) and the United States (right panel). China entered the scene as a destination market for plastic waste in the early 2000s, around the time

of its entry to the WTO. By the 2010s China was the dominant destination market for plastic waste. The amount of foreign-origin plastic waste bought by importers in China alone dwarfed amounts bought by importers in entire world regions. After 2017, the volume of plastic waste imported into China drops to effectively zero in 2018 and stays that way for the next years. A major takeaway from Figure 5 is that after 2017 there is a “missing bar” for each of the EU-27 and US – a quantity of metric tons of plastic waste that is not exported, but that pre-ban trends suggest would be exported. Given that the production of plastic waste is highly inelastic, it is not that this plastic waste suddenly did not exist in 2018, 2019, and 2020. It just was no longer being shipped abroad, so it does not show up in trade statistics. But by looking at trade statistics, we can infer the situation – suddenly, hundreds of thousands of metric tons of plastic waste had to be dealt with at home in EU member states and at home in the US.

Nonetheless, the patterns and sizes of the bars indicating exports to other destination regions reinforce trends discussed above – EU-27 exports involved neighbors in the Global North to a huge extent, not only after but even before the China garbage shock.⁹ In stark contrast, US plastic waste has consistently traveled to the Global South, which for well-explored reasons in myriad academic literatures are remote to US domestic politics.

5. Linking trade protection and policy responses

It is of course well understood that, compared to the US, the EU has a set of interests and institutions much more interested in environmental protection. The observable implication is that domestic politics in the EU is more conducive to deepening trade-environment linkages than in the US. Based on the data above, we expect that the quantitative fallout of the China garbage shock reinforced, rather than challenged, longstanding in regulatory and governance approaches between the EU and US. In what follows, we substantiate this claim by tracing discursive and policy responses in the EU and US in the years following the China garbage shock.

5.1 European Union discourse and policy

Waste management policies in the EU are influenced by multiple political forces at local and national levels, although action the supranational EU-level has been longstanding and wide-ranging. Textbox 1 summarizes key legislative and policy reforms at the EU level. Even before the China garbage shock, the EU already had supranational policies on packaging waste, waste management, plastic bags, and circular economy goals. At the same time, the number of waste management-relevant policies accelerated and proliferated after 2017. Of course, this does not mean that the European Commission and Parliament rushed to mobilize a regulatory arsenal in response to the China garbage shock. It would be naïve to overlook the research, coalition-building, bargaining, and time needed to bring these types of policies into effect. Still, the

⁹ There is evidence of diversion to different destination markets within world regions for both the EU and US, which is masked in Figure 5 for clarity's sake.

timeline does underpin that reform to waste management became the more pressing in the shock's aftermath.

Textbox 1: Key policy platforms and legislation in EU waste management

- 1994** Directive on Packaging and Packaging Waste (PPWD)
- 2006** Waste Shipments Regulation
- 2008** Waste Directive: Established framework for waste management in the EU.
- 2015** Plastic Bags Directive
- 2015** Circular Economy Action Plan
- 2018** Amendments to PPWD: Established new recycling targets for glass, plastic, paper and cardboard, metal, aluminum, and wood.
- 2018** European Strategy for Plastics in a Circular Economy
- 2018** Waste Directive Amendments: Addressed importance of critical raw materials, new recycling targets, separate collection requirements, and extended producer responsibility (EPR) schemes, among other topics.
- 2018** Landfill Directive: Prompted EU countries to implement national strategies to progressively reduce biodegradable municipal waste sent to landfills + reduce landfill waste to a maximum of 10% in 2035.
- 2019** Single Use Plastics Directive: Required members to ban a series of substitutable plastics.
- 2023** Waste Directive: Amendments
- 2024** Waste Shipments Regulation: Amendments
- 2025** Single Use Plastics Directive: Amendments

Sources: FAOLEX, EU Commission

Discursively, EU policymakers have not hesitated to point to China's actions as an impetus for policy responses. As an important example, the 2018 European Strategy for Plastics in a Circular Economy innovated on previous proposals by homing in on waste plastics – whose import was fully banned by China – as a priority for sustainability objectives. Though the report refers at times abstractly to the issue of “recent developments in international trade,” it is also explicit: “in particular China's recent announcements of its decision to ban import of certain types of plastic waste” (European Commission 2018, p. 8, fn 29). Policies discussed do not reject the principle of waste trade, but rather frame China's new import restrictions as an opportunity for EU recyclers to develop other markets for recycled plastics, including within the EU common market (ibid pp. 4, 9). Such discourse leans into EU trade policy reforms as in furtherance of environmental goals, not in opposition to them. Moreover, criticism of Chinese import bans is missing: EU discourse is about adapting, rather than fighting the closing of foreign markets to EU waste products.

In terms of policy output, waste plastic is at the center of 2018 amendments to the EU's Packaging and Packaging Waste Directive (PPWD), which set new recycling and material recovery targets.

One of the EU's most consequential actions has been the Single Use Plastics Directive (2019), which requires members to ban a series of substitutable plastics and has fundamentally changed consumers' experiences in the member states of the EU. Moreover, the acceleration of waste management reforms has not been confined to the plastics sector. In recent years, the EU has targeted new oversight and regulation of a broad range of waste categories including glass, paper and cardboard, metal, aluminum, wood, biodegradable municipal waste, and waste as a source of critical raw materials. Broader policies addressing a variety of waste streams – including products and categories not banned by China – signal that policy action on trade-environment linkages emerged beyond the confines of the initial shock.

The EU's regulatory offensive has been paired with new investments in research and development related to waste management. First, Horizon research grants (Horizon 2020 programs and their successor Horizon Europe) have supported several international research projects on waste, most notably marine plastic waste. Second, European Structural and Investment Funds in general, and the specific EU Cohesion Fund aimed at reducing socio-economic and territorial inequalities across the European Union, allocated over EUR 5.3 billion for improving waste management from 2014 to 2020. A stated goal is to increase waste recycling capacity in targeted countries (European Commission 2025).¹⁰ Third, the EU launched a Circular Economy Finance Support Platform designed to raise awareness among investors and facilitate access to finance for circular eco-projects (EC 2018; Xu, Zhang and Han, 2024).

While auditing the impact of these policies is outside our scope, there is evidence of positive trends. With regard to plastics, the industry association Plastics Europe has reported positive trends in plastic recycling rates: while in 2018 7.2% of the content of new products was recycled plastic (4 million tons), this number had risen to 12.6% (6.8 million tons) in 2022.¹¹ They also report upward trends in waste collection, recycling, and energy recovery from EU post-consumer plastics (Plastics Europe 2024). Admittedly, these positive trends must be contextualized with the persistent rise of EU post-consumer plastic waste in absolute terms, as well as household recycling rates that have stagnated or declined since the mid 2010s (Eurostat 2025). Still, the flurry of regulatory activism and enacted waste management reforms, as well as the absence of criticism of China's environmentally justified protectionist policies, illustrate multiple avenues through which motivated actors can capitalize on trade-environment linkages in domestic politics.

Further, the EU's regulatory innovations have not been confined to Europe. Specifically, the shock to plastic waste exports coincided with a renewal of diplomatic activism to regulate plastics internationally. First, in 2019, 187 countries agreed to amend the 1989 Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal to add contaminated and unrecyclable plastic waste, along with several other types of plastics, to the list of controlled substances. Throughout the drawn-out negotiation process, the EU had been

¹⁰ 15 European countries are eligible for EU Cohesion funds: Bulgaria, Croatia, Cyprus, Czechia, Estonia, Greece, Hungary, Latvia, Lithuania, Malta, Poland, Portugal, Romania, Slovakia and Slovenia.

¹¹ Plastics Europe represents 90% of Europe's polymers producers.

particularly active in both regional and multilateral forums, promoting key principles such as the precautionary principle and extended producer responsibility (Xu, Zhang and Han 2024). Enacted amendments are in line with the original Basel Convention, such that exporters are required to follow the Prior Informed Consent (PIC) procedure to obtain approval from importing countries before transporting these products. The EU played an important role in the negotiations by demanding that only the cleanest categories of plastic waste be exempt from the PIC procedure. According to an environmental group that closely followed the negotiations, “The EU is not only leading by example but taking its Plastics Strategy to the international level” (Environmental Investigation Agency 2019).

Second, a clear sign of the EU’s leading role in reforming international trade in waste is the fact that the EU’s own rules governing waste management now have a stricter framework than the Basel Convention. In 2024, the EU reformed its 2006 Waste Shipment Regulation (WSR), the framework regulating the transborder shipment and trade of waste within the EU and beyond its borders.¹² Key revisions include that plastic waste exports to non-OECD countries are banned starting November 2024, for at least 2.5 years. Thereafter, plastic waste shipments might be allowed again under stricter rules on the EU side and improved environmental management on the side of the importing country (EC DG Environment 2024). Additionally, the EU has committed to improving tracking efforts in order to crack down on illegal waste shipments. For DG Environment policy officer Yorg Aerts, while these efforts were associated with environmental policymakers, interagency dialogue brought in trade policymakers as well to factor the implications of the WSR for international trade.¹³

Third, the EU has also played an important role in the negotiations of an international, legally binding Plastics Treaty since March 2022. While the treaty was supposed to be finalized in Busan in December 2024, parties agreed to extend negotiations into 2025 given persistent disagreements over the financing of environmental measures, the classification of chemicals in plastics, and the adoption of upstream measures to reduce plastic production (European Union News 2024). For its part, the EU has advocated on the part of stricter provisions. It joined a group of 68 countries from all continents called the “High Ambition Coalition to End Plastic Pollution,” which is committed to legally binding measures to end plastic pollution by 2040 (EC DG Environment 2022). The EU has also acted as a broker in negotiations, building on overlapping interests in international action over climate, ocean protection, and waste management. In particular, the EU has sought to forge a compromise between advocates of a top-down approach with strict monitoring rules and defenders of a bottom-up solution based on voluntary reporting mechanisms (Xu, Zhang, and Han 2024).

Arguably, the EU’s embrace of circular economy principles has been the world’s most comprehensive (Barrowclough and Deere Birkbeck 2020). Discourse, policy proposals, and

¹² The WSR was first introduced in 1993 after the OECD adopted its own decision on the control of transboundary movement of waste. It was amended in 2006 as a result of adjustments to the Basel Convention, before being revised before its major reform in 2024. For more details, see Karamfilova (2021).

¹³ Videoconference with the authors, 16 June 2023.

enacted policies both within the EU and in the EU's efforts at the international level suggest that China's protectionism reinforced the EU's green ambitions.

5.2 The US response

Comparing the US and EU responses to the China garbage shock requires coming to grips with a paradox: in some policy spheres, the US federal system has a more decentralized regulatory regime than the European Union and its members. Environmental policy is one of these areas, in which the US Environmental Protection Agency (EPA) sits under the President's authority in the executive branch and has historically had a limited purview compared to environmental policymakers in the EU and elsewhere. In the realm of waste regulation, Congress has endowed the EPA with powers over specific sub-policy areas including waste-to-energy efforts, landfill oversight, and hazardous waste definitions. The 1976 Resource Conservation and Recovery Act (RCRA) and its amendments (1984, 1992, 1996) focuses the EPA's efforts on assisting states and localities with funding of their autonomous policies (EPA 2025).

The China garbage shock did precede new federal efforts to formalize and harmonize waste policy. In 2019, during the first Trump administration, the EPA released its National Framework for Advancing the US Recycling System. Notably, the framework emerged not from Congressional direction but rather from a Republican administration known for anti-environment positions. Discursively, the EPA framework addressed "reduced export markets for recycled materials," explaining that "Historically, some of the recycled materials generated in the United States have been exported internationally. However, changing international policies have limited the export of materials" (EPA 2019). Unlike the EU, which specified China's role in these changes directly, China was never mentioned by name. Further, the EPA offered that "domestic markets for recycled materials need to be strengthened in the United States" (ibid p. 10). It did not address concerns about the quality of (internally or externally) traded goods as did EU discourse. Still, the EPA promoted education and outreach on recycling, improving measurement, and enhancing materials management infrastructure, though without contradicting the leadership of state and local jurisdictions on such issues.

In a surprising sign of bipartisanship, the Biden administration's 2021 National Recycling Strategy built upon the Trump administration's agenda, calling for similar goals of expanding markets, improving materials collection and management infrastructure, and standardizing measurement (EPA 2021). The 2021 strategy further addressed reducing contamination and enhancing circularity, reminiscent of EU goals. Yet again, "changes to global trade" that are "further amplifying the need for new markets" was the oblique reference to the China garbage shock and its consequences. Here again, China was not mentioned by name.

The Biden administration innovated further, creating in 2023 an Interagency Policy Committee (IPC) on Plastic Pollution and a Circular Economy to coordinate efforts to combat plastic pollution among US federal agencies. Notably, the US Trade Representative and the Department of Commerce's International Trade Administration were involved, suggesting a gradual institutionalization of trade-environment linkages (White House 2024). Still, the only clear federal

policy enacted in the Biden administration is the Inflation Reduction Act's provision supporting food waste for energy recovery through anaerobic digestion systems (Levin 2022). Further moves toward policy coordination and reform have died out in the second Trump administration, which in its first months rolled back Biden-era environmental protections and has sought to undo rather than coexist with preexisting greener state and local efforts.

In Congress, reform efforts have been stymied. The Break Free From Plastic Pollution Act (BFFPPA) was an ambitious bill repeatedly introduced during the Biden administration in 2020, 2021, and 2023 – under either full Democratic or split control – but it did not advance even to committee review. The bill echoed EU efforts aimed at reducing plastic production, prohibiting additional toxic substances, and innovating on recycling and materials recovery. The BFFPPA faced fierce opposition from the petrochemical industry, including the American Chemistry Council, the Plastic Industry Association and the American Fuel and Petrochemical Manufacturers (AFPM 2021, Rachal 2023). After Trump's reelection, the plastics industry introduced what critics called a "wish list for a second Trump administration," endorsed by the same groups (Song 2024). Although there are influential waste and recycling industry associations in both the EU and US, the breadth and power of the lobbying coalition in the US illustrates how countermobilization can perpetuate trade disconnects (Velut 2023).

Internationally, no US administration has seriously considered joining the 1989 Basel Convention, though its otherwise near-universal membership means that Basel provisions influence EPA policy. The EPA noted that 2019 adoption of plastics amendments to the Basel Convention, "while limiting US export markets for recyclable material, offer new incentives to develop domestic market opportunities." That increased internal trade might address new limits on international trade has become a theme of US responses to heightened global regulation. Regarding the Global Plastics Treaty, the Biden administration delegations had been weakly supportive but also noncommittal, in stark contrast to the EU's enthusiasm. The role of petrochemical regulatory capture is on display when, for example, the US and other oil producers like Saudi Arabia repeat parallel objections during intergovernmental negotiation committee meetings. Despite eschewing a variety of international environmental fora, the second Trump administration did send a delegation to informal talks in summer 2025. However, the US official position is now fully against any plastic production limits, and negotiators are uncertain as to the US's intentions – "they could be willing to do something, or let others do their thing and not ratify... But if they want to hinder others, it will be very challenging" (cited in Civillini 2025).

In short, the China garbage shock was followed by attention to changes in international markets, under both Republican and Democratic administrations. Yet, the shock has not proven strong enough to catalyze direct discourse or policy follow-through on the part of the US, whether domestically or internationally.

6. Conclusion

The China garbage shock of 2017 upended global markets for important waste products, translating into trade declines and diversion that disrupted the NIMBY waste export status quo for both the European Union and the United States. Empirically, we demonstrate that subsequent trade diversion meant replacement markets were closer to home in the EU, geographically and socio-economically. In contrast, the US replacement export markets remained at great distances and lower socio-economic status. This variation in visibility reinforced what are already well-known differences between EU green ambitions and US reluctance. While in the US both Republican and Democratic administrations have acknowledged the disruption, albeit obliquely, regulatory proposals and enacted reforms are nowhere near the longstanding EU efforts to regulate and link waste management and trade policies.

The fact that the China garbage shock reinforced the EU-US divergence on both domestic and international fronts may not seem surprising. What, then, is there to learn from this deep dive into the potential green virtues of Chinese protectionism and the impact of external shocks on trade linkages? First, it is important to establish the regular workings of classic political economy forces in an underexplored substantive domain. The data and narratives above suggest many theoretical mechanisms at work – political visibility, federalism, regulatory capture, path dependency, collective action problems, and more. Thus, a robust body of theory is consistent with our findings that, yes, the EU provides proof of concept that Global South protectionism – and not just trade liberalization in the form of FTA provisions – can push forward trade-environment linkages in the Global North, as probably a most-likely case.

Second, this Global South action was nowhere near enough to upend policy outcomes in the US, which from political economy first principles is probably a least likely case. Still, US exporters have had to adjust and trade patterns are demonstrably different. As such, new sets of “winners” and “losers” are emerging in local constituencies, which suggests local experimentation with new policies even if the federal level remains stuck. Indeed, in 2021 the US National League of Cities issued a comprehensive policy framework, “Beyond Recycling: Policy to Achieve Circular Waste Management,” which discusses China’s National Sword Policy by name and spells out best practices to adjust in its wake (NLC 2021, Rosengren et al 2019). From the point of view of the US, a major competitor’s supra-national policy and its subnational units’ municipal policies are pushing in the same direction. Might this be enough to break through ossified national politics? Multiscalar synergies are increasingly recognized in studies of climate politics; adding waste management to the mix will allow us to better assess how externally valid climate coalitions are to other issue areas (e.g. Gulzar et al 2024, Rudra et al 2018, Stern 2004). Future research could, therefore, zoom in on the role that local actors play in reinforcing or obstructing trade-waste management linkages, building upon the emerging literature on subnational trade politics (e.g. Broschek & Goff 2020, Lequesne & Paquin 2017, Schiavon, 2020).

Third, a comparative study of contemporary adjustment in the EU and US only scratches the surface of the domestic and international politics of the global waste trade in the Global North, and it has little to say about its development and consequences in the Global South. Waste production is extremely inelastic. The rhetoric and policy in states reliant on export of never-ending waste products sheds light on the options available to Global South importing states,

which both benefit economically and suffer environmentally from the waste trade (Wellhausen 2025). Indeed, even as China made the radical decision of banning imports of 26 specific waste products, it did not ban the import of all waste products. Concerns are rising over its material and normative consequences, and still the waste trade continues. What economic policies offer linkage opportunities for green virtues, in not just most- but also least-likely places? These questions can no longer stay hidden in the black trash bag.

References

Aklin, Michael. 2016. "Re-exploring the Trade and Environment Nexus Through the Diffusion of Pollution." *Environmental and Resource Economics* 64(4):663–682.

American Fuel and Petrochemical Manufacturers (AFPM). 29 March 2021. "AFPM Statement on the Break Free from Plastic Pollution Act." States News Service.

Autor, David, David Dorn, and Gordon Hanson. 2016. "The China Shock: Learning from Labor-Market Adjustment to Large Changes in Trade." *Annual Review of Economics* 8:205–240.

Barrowclough, Diana and Carolyn Deere Birkbeck. April 2020. "Transforming the Global Plastics Economy: The Political Economy and Governance of Plastics Production and Pollution." Global Economic Governance Programme. GEG Working Paper 142.

Basel Convention. (n.d.) "Questions and answers related to the Basel Convention Plastic Waste Amendments." Available from: <https://www.basel.int/implementation/plasticwaste/plasticwasteamendments/faqs/tabid/8427/default.aspx> (last accessed July 31, 2025).

Bastiaens, Ida, and Evgeny Postnikov. 2017. "Greening Up: The Effects of Environmental Standards in EU and US Trade Agreements." *Environmental Politics* 26 (5): 847–869.

Bernauer, Thomas. 2013. "Climate Change Politics." *Annual Review of Political Science* 16(1):421–448.

Brandi, Clara, Dominique Blumer, and Jean Frederic Morin. 2019. "When do international treaties matter for domestic environmental legislation?" *Global Environmental Politics* 19(4):14–44.

Brandi, Clara, and Jean Frederic Morin. 2023. *Trade and the Environment. Drivers and Effects of Environmental Provisions in Trade Agreements*. Cambridge University Press.

Brooks, Amy, Shunli Wang and Jenna Jambeck. 2018. "The Chinese Import Ban and its Impact on Global Plastic Waste Trade." *Science Advances* 4(6):1–8.

Broschek, J., & Goff, P. (ed.) (2020). *The Multilevel Politics of Trade*. University of Toronto Press.

Civillini, Matteo. 4 July 2025. "US comes out against plastic production limits in UN treaty at deadlocked talks." *Climate Home News*.

Colgan, Jeff D. and Miriam Hinthorn. 2023. "International Energy Politics in an Age of Climate Change." *Annual Review of Political Science* 26(1):79—96.

Cotta, Benedetta. 2020. "What Goes Around, Comes Around? Access and allocation problems in Global North-South waste trade." *International Environmental Agreements* 20(2):255–269.

De Ville, Ferdi, Simon Happersberger, and Harri Kalimo. 2023. "The Unilateral Turn in EU Trade Policy? The origins and characteristics of the EU's new trade instruments." *European Foreign Affairs Review* 28:15—34.

Environmental Investigation Agency. 10 May 2019. "Basel Convention scores a major victory in the war on plastic pollution," Available from: <https://eia-international.org/news/basel-convention-scores-a-major-victory-in-the-war-on-plastic-pollution/> (last accessed July 31, 2025).

Environmental Protection Agency (EPA). November 2019. "National Framework for Advancing the US Recycling System." p 2. Available from: https://www.epa.gov/sites/default/files/2019-11/documents/national_framework.pdf (last accessed July 31, 2025).

EPA. November 2019. "National Framework for Advancing the US Recycling System." p 2. Available from: https://www.epa.gov/sites/default/files/2019-11/documents/national_framework.pdf (last accessed July 31, 2025).

EPA. 11 February 2025. "History of the Resource Conservation and Recovery Act (RCRA)." Available from: <https://www.epa.gov/rcra/history-resource-conservation-and-recovery-act-rcra> (last accessed July 31, 2025).

European Commission (EC). 2018. "A European Strategy for Plastics in a Circular Economy: Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the committee of the Regions." Brussels, 16 January.

EC. 2025. "Cohesion Open Data Platform, 2014-2020," available from: https://cohesiondata.ec.europa.eu/cohesion_overview/14-20 (last accessed July 31, 2025).

EC Directorate General for Environment. 4 November 2022. "Advancing towards Zero-Pollution: EU joins call for ambitious global agreement to end plastic pollution by 2040." Available from: <https://environment.ec.europa.eu/news/advancing-towards-zero-pollution-eu-joins-call->

[ambitious-global-agreement-end-plastic-pollution-2040-2022-11-24_en](#) (last accessed July 31, 2025).

EC Directorate General for Environment. 20 May 2024. "New Regulation on waste shipments enters into force." Available from: https://environment.ec.europa.eu/news/new-regulation-waste-shipments-enters-force-2024-05-20_en (last accessed July 31, 2025).

European Union News. 11 December 2024. "Industry, Government and Conservationists in the Dark as UN Plastics Treaty Falters for a Fifth Time."

Eurostat. 2025. "Packaging waste by waste management operations" (env_waspac database), available from: https://ec.europa.eu/eurostat/databrowser/view/env_waspac_custom_15449979/default/table?lang=en (last accessed February 15, 2025).

Favarin, Serena and Alberto Aziani. 2020. "Global Waste Trafficking and Its Correlates." *Journal of Contemporary Criminal Justice* 36(3):351–383.

Gaulier, Guillaume and Soledad Zignago. 2010. *BACI: International Trade Database at the Product Level*. Working Papers 2010-23 CEPII.

George, Clive & Shunta Yamaguchi. 2018. "Assessing Implementation of Environmental Provisions in Regional Trade Agreements." OECD Trade and Environment Working Papers 2018/01, available from: <https://doi.org/10.1787/18166881> (last accessed July 31, 2025).

Goldstein, Joshua. 2020. *Remains of the Everyday: A century of recycling in Beijing*. University of California Press.

Gulzar, Saad, Apoorva Lal, and Benjamin Pasquale. 2024. "Representation and Forest Conservation: Evidence from India's scheduled areas." *American Political Science Review* 118(2):764—783.

Jinnah, Sikina and Jean-Frédéric Morin (2020). *Greening Through Trade. How American Trade Policy Is Linked to Environmental Protection Abroad*. Cambridge, The MIT Press.

Karamfilova, Ekaterina. April 2021. "Waste Shipment Regulation Revision of Regulation (EC) No 1013/2006 on shipments of waste," European Parliamentary Research Service Briefing.

Kaza, Silpa et al 2018. "What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050." World Bank.

Kellenberg, Derek. 2015. "The Economics of the International Trade of Waste." *Annual Review of Resource Economics* 7(1):109—125.

- Kim, In Song, Steven Liao, and Kosuke Imai. 2020. "Measuring Trade Profile with Granular Product-Level Data." *American Journal of Political Science* 64(1):102–117.
- Lebreton, L. et al 2018. "Evidence that the Great Pacific Garbage Patch is rapidly accumulating plastic." *Nature: Scientific Reports* 8(4666).
- Lepawsky, Josh. 2018. *Reassembling Rubbish: Worlding Electronic Waste*. MIT Press.
- Lequesne, Christian, & Stéphane Paquin. 2017. "Federalism, Paradiplomacy and Foreign Policy: A Case of Mutual Neglect." *International Negotiation*, 22(1), 183-204.
- Levin, Michael H. 22 August 2022. "The IRA Revolutionizes AD Tax Credits." BioCycle.
- Liboiron, Max. 2021. *Pollution is Colonialism*. Duke University Press.
- Ma, Zijie et al 2021. "Material Flow Patterns of the Global Waste Paper Trade and Potential Impacts of China's Import Ban." *Environmental Science and Technology* 55(13):8492–8501.
- McKee, D. L. 1996. "Some Reflections on the International Waste Trade and Emerging Nations." *International Journal of Social Economics* 23(4-6):235–244.
- Minter, Adam. 2015. *Junkyard Planet: Travels in the billion-dollar trash trade*. Bloomsbury USA.
- Morin, Jean-Frédéric, Clara Brandi, and Jakob Schwab. 2024. "Environmental agreements as clubs: Evidence from a new dataset of trade provisions." *The Review of International Organizations* 19(1):33-62.
- Morin, Jean-Frederic, Andreas Dür and Lisa Lechner. 2018. "Mapping the Trade and Environment Nexus: Insights from a new dataset." *Global Environmental Politics* 18(1):122–139.
- National League of Cities. November 12, 2021. "Beyond Recycling: Policy to Achieve Circular Waste Management." Available from: <https://www.nlc.org/resource/beyond-recycling-policy-to-achieve-circular-waste-management/> (last accessed July 31, 2025).
- Okafor-Yarwood, Ifesinachi and Ibukun Jacob Adewumi. 2020. "Toxic Waste Dumping in the Global South as a Form of Environmental Racism: Evidence from the Gulf of Guinea." *African Studies* 79:285–304.
- O'Neill, Kate. *Waste Trading Among Rich Nations: Building a new theory of environmental regulation*. MIT Press, 2000.
- O'Neill, Kate. 2019. *Waste*. Polity.

Pacini, Henrique et al 2021. "Network Analysis of International Trade in Plastic Scrap." *Sustainable Production and Consumption* 27:203–216.

Plastics Europe. 2024. "The Circular Economy for Plastics." Available from: <https://plasticseurope.org/knowledge-hub/the-circular-economy-for-plastics-a-european-analysis-2024/>

Rachal, Maria. 26 October 2023. "Amped-up 'Break Free From Plastic' legislation reintroduced in Congress." Packaging Dive.

Roberts, J. Timmons, Bradley C. Parks, and Alexis A. Vásquez. 2004. "Who Ratifies Environmental Treaties and Why? Institutionalism, Structuralism and Participation by 192 Nations in 22 Treaties." *Global Environmental Politics* 4(3):22–64.

Rosengren, Cole et al 15 November 2019. "How recycling has changed in all 50 states." Waste Dive.

Rudra, Nita, Meir Alkon, and Siddharth Joshi. 2018. "FDI, Poverty, and the Politics of Potable Water Access." *Economics & Politics* 30(3):366—393.

Schiavon, Jorge. 2020. *Comparative Paradiplomacy*. Routledge.

Song, Lisa. 23 October 2024. "The Plastics Industry's Wish List for a Second Trump Administration." ProPublica.

Stern, David. 2004. "The Rise and Fall of the Environmental Kuznets Curve." *World Development* 32(8):1419–1439.

Torrente-Velásquez, Jorge M., et al 2020. "A Waste Lexicon to Negotiate Extended Producer Responsibility in Free Trade Agreements." *Resources, Conservation and Recycling* 156(104711).

Tran, Trang, Hiromasa Goto, and Takuma Matsuda. 2021. "The Impact of China's Tightening Environmental Regulations on International Waste Trade and Logistics." *Sustainability* 13(2):1—14.

Velut, Jean-Baptiste. 2022. "Environmental Allies and Trade Competitors: A comparative analysis of US and EU governance models for the trade-and-climate nexus." In E. Fahey and I. Mancini (eds.), *Understanding the EU as a 'Good' Global Governance Actor*. Edward Elgar. 74—90.

Velut, Jean-Baptiste et al with LSE Consulting. 2022. "Comparative analysis of trade and sustainable development provisions in trade agreements." LSE Consulting. Available from:

https://trade.ec.europa.eu/doclib/docs/2022/february/tradoc_160043.pdf

Velut, Jean-Baptiste. 2023. “Trade Linkages or Disconnects? Labor Rights and Data Privacy in US Digital Trade Policy.” *Politics and Governance* 11(1):249—260.

Wellhausen, Rachel L. 2023. “Waste Not, Want Not.” Working paper. University of California Institute on Global Conflict and Cooperation.

Wellhausen, Rachel L. 2025. “Tariffs as Environmental Protection: Evidence from the Global South after the China garbage shock.” Working paper.

White House. July 2024. “Mobilizing Federal Action on Plastic Pollution: Progress, Principles, and Priorities.” Available from: <https://bidenwhitehouse.archives.gov/wp-content/uploads/2024/07/Mobilizing-Federal-Action-on-Plastic-Pollution-Progress-Principles-and-Priorities-July-2024.pdf> (last accessed July 31, 2025).

Yamaguchi, Shunta. 2021. “International Trade and Circular Economy: Policy Alignment.” OECD Trade and Environment Working Papers No. 2021/02. OECD Publishing, Paris.

Yang, Shiming. 2020. “Trade for the Environment: Transboundary hazardous waste movements after the Basel Convention.” *Review of Policy Research* 37(5):713–738.

APPENDIX

A1. Full waste product list

050100 050210 050290 050300 050510 050590 050690 050710 050790 050800 150200 180200
 230210 230220 230230 230240 230250 230310 230320 230330 230400 230500 230610 230620
 230630 230640 230641 230649 230650 230660 230670 230690 230800 230810 230890 240130
 251720 252530 261900 262011 262019 262020 262021 262029 262030 262040 262050 262060
 262090 262091 262099 262100 262110 262190 271390 284440 284450 300680 300692 380400
 382510 382520 382530 382541 382549 382550 382561 382569 382590 391510 391520 391530
 391590 400400 401220 401700 410110 410120 410121 410122 410129 410130 410140 410150
 410210 410221 410229 410310 410330 410390 411000 411520 440130 440131 440139 440140
 450190 470710 470720 470730 470790 500300 500310 500390 510310 510320 510330 510400
 520210 520291 520299 530130 530290 530390 530490 530500 530519 530529 530590 530599
 550510 550520 630900 631010 631090 700100 711210 711220 711230 711290 711291 711292
 711299 720410 720421 720429 720430 720441 720449 740400 750300 760200 780200 790200
 800200 810110 810191 810197 810210 810291 810297 810310 810330 810420 810510 810530
 810600 810710 810730 810810 810830 810910 810930 811000 811020 811100 811211 811213
 811220 811222 811230 811240 811252 811259 811291 811292 811300 854810 890800

Notes: This is a list of 179 6-digit HS codes for waste and scrap products per HS code revisions 1992, 1995, 2002, 2007, 2012, and/or 2017. In general: a waste product is a traded product that contains waste, meaning it has a component with no further reuse value, with the implication that the leftover component is not re-exported and remains in the territory where the importing firm is located. To be classified as a waste product, the definition includes the term waste or scrap;¹⁴ it is a residual or byproduct from primary production processes;¹⁵ and/or the product is a one-time primary good intended to be processed into inputs for further use.¹⁶ 65 Consistent with industry practice, waste products are categorized into types, based on their origin: animal, chemical, metal, mineral, paper, plastic, textile, or vegetable. For more on type categorizations, full definitions, and relevant HS revisions, please contact the authors.¹⁷

¹⁴ For example, HS 470730: Paper or paperboard: waste and scrap, of paper or paperboard made mainly of mechanical pulp (eg newspapers, journals and similar printed matter).

¹⁵ For example, HS 382530: Residual products of the chemical or allied industries, not elsewhere specified or included: clinical waste.

¹⁶ For example, HS 401220: Rubber: used pneumatic tires. In contrast, used goods that are directly intended for resale – such as used vehicles in working order – are not waste products.

¹⁷ The count of codes by HS revision is as follows: 1992 HS revision - 129 waste products; 1996 - 131; 2002 - maximum of 144; 2007 - 134; 2012 - 134; 2017 - 135.